INCH-POUND

MIL-C-48635 (AR)
AMENDMENT 1
27 May 1993

MILITARY SPECIFICATION

CARTRIDGE, 81MM, PRACTICE M879
LOADING, ASSEMBLING, AND PACKING

This amendment forms a part of MIL-C-48635 (AR), dated 2 July 1987, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2, Under drawings delete: drawing numbers 9362798 and 9381730 and their subsequent titles and substitute:

"12630601 - Packing and Marking for Box, Wirebound, for Cartridge, 81MM, M879, M821E1 and M889E1

12630600 - Packing and Marking for Container, PA 149, Ammunition Fiber for Cartridge, 81MM M821E1, M889E1 and M879"

Insert the following before "(Copies of specifications...)"

"TEST OPERATING PROCEDURES

TECOM TOP 3-2-825 Location of Impact or Airburst Positions"

PAGE 5

TABLE I PROVING GROUND TEST LINE, delete "4.4.2.4" and substitute "4.4.3.1.1(A)"

Add note "A":

"A. A random sample of 96 cartridges shall be x-rayed and then sent to a proving ground for ballistic testing in accordance with 4.5.1."

PAGE 9

4.4.2.3, Delete under drawing number: "9381748" and substitute "12561559"

Delete under Next Higher Assembly: "9381747" and substitute "12630600"

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PAGE 10

4.4.2.3.1, Delete under Next Higher Assembly: "9130340" and substitute "12630600"

PAGE 12

4.4.2.4, Delete Major defects 113, 114 and 115.

Add new paragraph as follows:

"4.4.2.8 Cartridge, Practice, 81MM, M879 W/Fuze Practice, M751 (After Installation of Prop. Charge Sup. Ass'y) drawing number 9310340."

	Defects	AQL	Paragraph	Method
Critical:	None defined			
Major				
101	Support missing or damaged	.40%	3.1	Visual
102	Support tape missing	.40%	3.1	Visual
103	Tape pull tab inadequate	.40%	3.1	Visual
104	Warning label tape covers prop. chg. support tape pull tab	.40%	3.1	Visual
Minor				
201	Warning label damaged	.65%	3.1	Visual
202	Poor workmanship	.65%	3.1	Visual

PAGE 13

4.4.2.5, Delete title and substitute:

"Packing and Marking for Container, PA 149, Ammunition for Fiber Cartridge, 81MM M821E1, M889E1 and M879 (while packing)"

Delete under drawing number: "981730" and substitute "12630600"

Add under Next Higher Assembly: "1263601"

Under Special A, after "Packing" delete "Stop" and substitute "Clip".

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PAGE 14

4.4.2.6, Delete Title and substitute:

"Packing and Marking for Container, PA 149, Ammunition Fiber for Cartridge, 81MM M821E1 and M879 (after packing)"

Delete under drawing number "9381730" and substitute "12630600"

Delete under Next Higher Assembly "9362798" and substitute "12630601".

Major 102, Delete callout under Examination or Test and substitute:

"Sealing strip (fiberglass) incomplete or badly wrinkled (before applying additional sealing strip (cloth)"

Major 103, Delete callout under Examination or Test and substitute:

"Sealing strip (cloth) incomplete or badly wrinkled"

PAGE 15

4.4.2.7, Delete title and substitute:

"Packing and Marking for Box, Wirebound, for Cartridge, 81MM M879, M821E1 and M889E1"

Delete under drawing number "9362798" and substitute "12630361"

Minor 204, Delete in its entirety.

Add the following Minor Defect, Examinations While Packing (Before Sealing Box):

"207 Box containing plastic sheet not appropriately marked ...0.65% ...3.1 ...Visual

208 Graphical firing scale not present in every third box ...0.65% ...3.1 ...Visual

209 Box containing graphical firing scale not appropriately marked ...0.65% ...3.1 ...Visual

210 Plastic sheet not present in every sixth box ...0.65% ...3.1 ...Visual"

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Change from:

"4.4.3.1.1 Sampling plan. Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, ninety six (96) random sample cartridges shall be selected from each lot for this test. The lot shall be rejected if:

a. Two or more misfires occur.
b. One or more short rounds occur.
c. One or more metal parts separations occur.
d. Horizontal range standard deviation at charge 1 exceeds 30 meters or horizontal range standard deviation at charge 4 exceeds 83 meters.
e. Three or more duds occur."

Change to:

"4.4.3.1.1 Sampling plan. Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, ninety-six (96) random sample cartridges shall be selected from each lot for this test. The lot shall be rejected if:

a. Two or more misfires occur.
b. One or more short rounds occur.
c. One or more metal parts separations occur.
d. Horizontal range standard deviation at charge 0 exceeds 22 meters or horizontal range standard deviation at charge 4 exceeds 83 meters.
e. Three or more duds accumulatively occur in the charge 1 and charge 4 firings."

Change from:

"4.4.3.1.2 Modified sampling plan. After three consecutive lots have complied with the criteria of 4.4.3.1.1, forty eight random sample cartridges shall be selected from each lot for this test. The lot shall be rejected if:

a. Two or more misfires occur.
b. One or more metal parts separations occur.
c. Horizontal range standard deviations at charge 1 exceeds 30 meters or horizontal range standard deviation at charge 4 exceeds 83 meters.
d. Three or more duds occur."

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Change to:

"4.4.3.1.2 Modified sampling plan. After three consecutive lots have complied with the criteria of 4.4.3.1.1, sixty (60) random sample cartridges shall be selected from each lot for this test. The lot shall be rejected if:

a. Two or more misfires occur.
b. One or more short rounds occur.
c. One or more metal parts separations occur.
d. Horizontal range standard deviation at charge 0 exceeds 22 meters or horizontal range standard deviation at charge 4 exceeds 83 meters.
e. Three or more duds accumulatively occur in the charge 1 and charge 4 firings."

PAGE 17

Change from:

"4.5.1 Proving ground test. The test shall be performed at a Government Proving Ground in accordance with the applicable USATECOM Acceptance Test Procedure.

The cartridges shall be fired from a standard 81MM M252 mortar at an elevation of forty-five (45) degrees. Fire all rounds at 70 degrees F plus or minus 5 degrees F after preconditioning for twenty-four hours. One half of the sample cartridges shall be fired at charge one and the other half at charge four. The individual velocities, pressures, deflections, and ranges shall be recorded on the firing record. Observation shall be made for misfires and duds, and shall be recorded. In the event of a misfire, the ignition cartridge shall be removed and forwarded to ARDEC, ATTN: SMCAR-LCE-E for disassembly. Misfired ignition cartridges shall be not be disassembled by the proving ground. Should metal parts separation occur, a photograph shall be taken of the evidence and attached to the firing record. The velocity and range standard deviations and arithmetic averages shall be calculated and recorded. Temperature of the day, weather and the time of firing shall be recorded. Range probable error shall be calculated and recorded.

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Hawk or equivalent radar shall be used to observe for metal parts separation, unobserved short rounds and unobserved duds. In these events, an analysis should be made of the radar data and should be noted on the firing record. This shall include a comparison of times of flight of the unobserved rounds with that of the normal rounds. Prints of the radar traces of these rounds and prints of the preceding and succeeding rounds, shall be attached to the firing record. Data from ballistically acceptable lots may be discarded. A smear camera shall be used to photograph all samples. The camera shall be positioned to pick up each round just outside the second velocity coil. Should an unobserved dud, short round or any other abnormal performance occur, a photo print of these rounds, and prints of all other rounds in that particular charge zone in that lot, shall be attached to the firing record. Film from acceptable lots may be discarded."

Change to:

"4.5.1 Proving ground test. The test shall be performed at a Government Proving Ground in accordance with the applicable USATECOM Acceptance Test Procedure.

The cartridges shall be fired from a standard 81MM M252 mortar at an elevation of forty-five (45) degrees and shall be temperature preconditioned to 70 degrees F plus or minus 5 degrees F for a minimum of twenty-four hours prior to firing. Twenty-four (24) cartridges (or twenty (20 cartridges for the modified test quantity) shall be fired at charge 0, twenty-four (24) cartridges (or twenty (20) cartridges for the modified test quantity) shall be fired at charge 1, and the remaining forty-eight cartridges (twenty (20) cartridges for the modified test quantity) shall be fired at charge 4. The charge 4 phase shall be fired before the charge 1 phase. Sixteen (16) data points are required to validate the charge 4 firing phase. If need be, up to four (4) additional cartridges may be diverted from the charge 1 phase test samples to the charge 4 test phase so that the minimum 16 data point requirement can be met. Individual velocities, pressures, times of flight, deflections, and ranges shall be recorded on the firing record for all charge phases (i.e. charge 0, charge 1 and charge 4). Duds for the charge 0 firings shall be recorded for information only. Range and deflection data for the charge 1 firings shall be recorded for information only. Observation shall be made for misfires and duds in all test phases and shall be recorded. The charge 0 phase shall be shot on a clean field. Each charge 0 impacted

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round shall be flagged and measured directly by optical methods. For both the charge 1 and charge 4 firing phases, three downrange observers are required in the acquisition of range and deflection data. These observers shall be positioned with at least one beyond the impact zone and at least one short of the impact zone in such a manner that their downrange data can be tabulated in accordance with the procedures defined in TECOM TOP 3-2-825, Appendix C. The fourth observer as referred to in the TOP is desired, but not required for the charge 1 and charge 4 firing phases.

After the first five (5) test rounds have been fired, the round impact data taken by the observers shall be used to validate their accuracy. If the reported data cannot be validated per the TOP, then those data shall be recorded, but not used in the downrange performance calculations. The observers shall then be required to analyze and correct the nature of their errors before firing can resume. In the event of a misfire, the ignition cartridge shall be removed and forwarded to ARDEC, ATTN: SMCAR-FSA-MM for disassembly. Misfired ignition cartridges shall not be disassembled at the proving ground. Should metal parts separation occur, a photograph shall be taken of the evidence and attached to the firing record. The muzzle velocities, range standard deviations and arithmetic averages for all rounds shall be calculated and recorded. Temperature of the day, weather conditions and the time of firing shall be recorded. Range probable error shall be calculated and recorded for all firing phases.

A smear camera shall be positioned to record each round in flight at about twenty (20) feet out of the muzzle. Should an unobserved dud, short round or any other abnormal performance occur, a photo print of these rounds, and prints of all other rounds in that particular charge zone in that lot, shall be attached to the firing record. Film from acceptable lots may be discarded."

PAGE 20

Paragraph 6.5, last two lines: Change from "Forty-eight samples thereafter," to "Sixty samples thereafter,"

Custodian:
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Preparing activity:
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(Project 1315-AD35)

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